📄 Resume Management System – Full Stack Documentation

1. Project Overview

The Resume Management System is a full-stack application designed to manage resume files efficiently. It allows users to upload, store, view, download, and delete resumes.

The system is built using:
Backend: Spring Boot (Java 17)
Frontend: Angular 18
Database: MongoDB
Storage: Local File System
The architecture follows a clean separation of concerns with scalable design principles.

2. Architecture

🔹 High-Level Architecture

Client (Angular UI)
↓
REST API (Spring Boot)
↓
Service Layer (Business Logic)
↓
Repository Layer
↓
MongoDB (Metadata Storage)

Service Layer → Local File System (Resume Files)

🔹 Frontend Architecture (Angular)

Standalone Components
Feature-based folder structure

```
src/app/
│
├── core/services → API calls
├── features/resume
│   ├── upload → Upload UI
│   ├── list → Resume list UI
│
├── shared/models → Data models
```

3. Backend (Spring Boot)

🔹 Responsibilities

Handle file upload/download
Validate file type & size
Store files in local system
Store metadata in MongoDB
Provide REST APIs

🔹 Features Implemented

✔ Upload Resume (PDF/DOC/DOCX)
✔ Store file in local system
✔ Store metadata in MongoDB
✔ Fetch all resumes
✔ Download resume
✔ Delete resume (file + DB)

🔹 Database Design

Database: biz
Collection: resumes
Sample Document:

```
{
"_id": "123",
"fileName": "resume.pdf",
"fileType": "application/pdf",
```

"filePath": "/path/to/file",
"fileSize": 70965,
"uploadedAt": "2026-03-25T02:12:58"
}

🔹 File Storage
Stored in:
~/workspace/docs/resume/

Unique naming using UUID to prevent overwrite

4. Frontend (Angular)
🔹 Responsibilities
Upload file via UI
Show resume list
Preview (PDF only)
Download & delete resumes
Provide validation & feedback

🔹 Features Implemented
✔ Upload UI with validation
✔ File type & size validation
✔ Toast notifications (auto-hide)
✔ Loading spinner during upload
✔ Resume listing table
✔ Empty state UI
✔ File type badges (PDF/DOC/DOCX)
✔ Preview (PDF only)
✔ Download functionality
✔ Delete with confirmation
✔ Disabled preview for non-PDF

🔹 UI/UX Enhancements

Bootstrap-based responsive design
Clean navbar and layout
Disabled upload button for invalid files
Error handling with user-friendly messages

5. API Contracts

🔹 Upload Resume

POST /api/resumes/upload

Request: form-data (file)
Response: Resume metadata

🔹 Get All Resumes

GET /api/resumes

🔹 Download Resume

GET /api/resumes/download/{id}

Response: File (Blob)

🔹 Delete Resume

DELETE /api/resumes/{id}

6. Data Flow

🔹 Upload Flow

User selects file in UI
Frontend validates file
API call to backend
Backend validates again
File saved to system
Metadata stored in MongoDB
Response returned

- Download / Preview Flow

User clicks preview/download
Request sent to backend
Backend fetches metadata
File read from system
Returned as Blob
Opened in browser (preview) or downloaded

- Delete Flow

User clicks delete
Confirmation shown
API call sent
File deleted from system
Record removed from MongoDB
UI refreshed

7. Setup Instructions

- Backend Setup

Clone project
Configure MongoDB:
spring.data.mongodb.uri=mongodb://localhost:27017/biz

Set file path:
file.upload.dir=~/workspace/docs/resume/

Run:
mvn spring-boot:run

- Frontend Setup

Install dependencies:
npm install

Run Angular:
ng serve -o

8. Future Enhancements

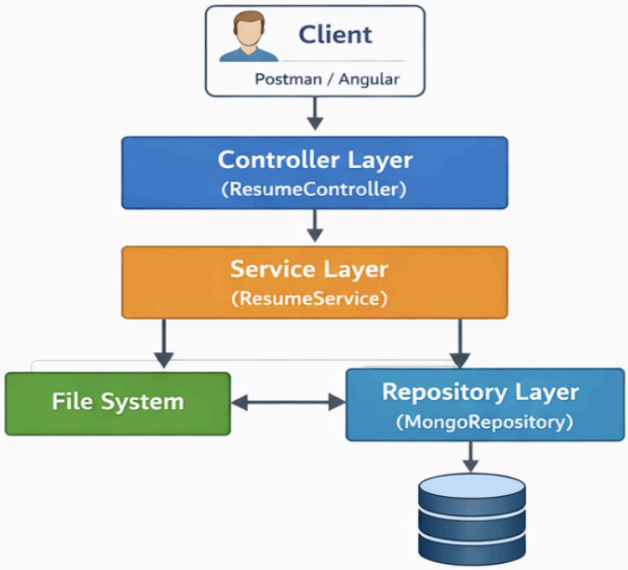
- ☐ JWT Authentication & Authorization
- ☐ Role-based access control
- ☐ AWS S3 integration (cloud storage)
- ☐ Resume search & filtering
- ☐ Pagination
- ☐ Drag & drop upload
- ☐ AI-based resume parsing

9. Conclusion

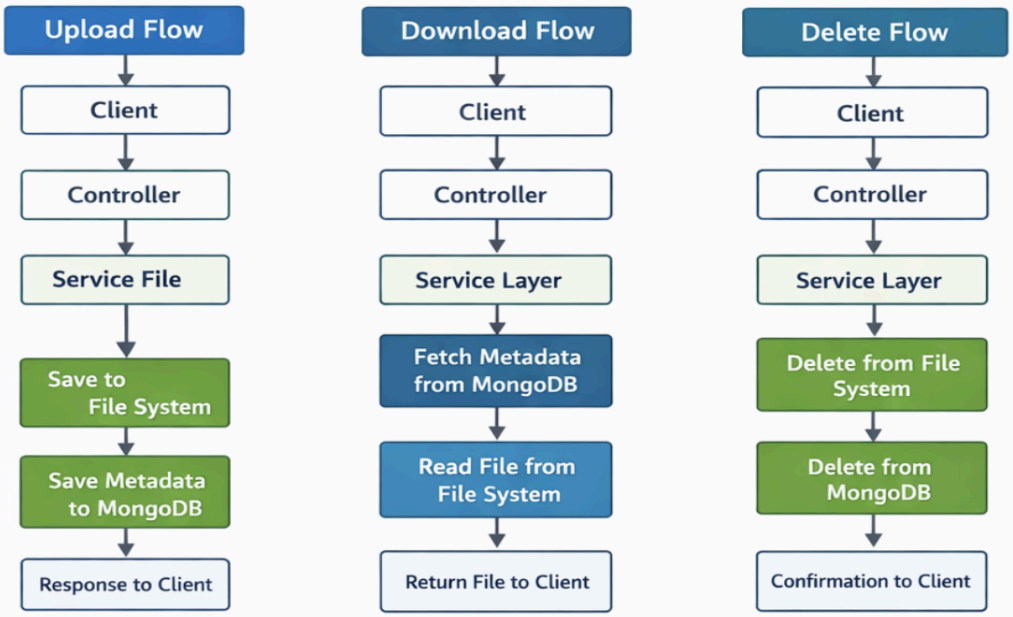
The Resume Management System is a fully functional full-stack application demonstrating clean architecture, proper validation, and modern UI/UX practices.

It handles the complete lifecycle of resume files and is scalable for real-world production enhancements.

Architecture



Data Flows



📄 Backend Dockerization Documentation

1. Overview
The backend of the Resume Project is developed using Spring Boot and packaged as an executable JAR file. To ensure portability and consistency across environments, the backend is containerized using Docker.

2. Prerequisites
Before Dockerizing the backend, the following requirements must be met:

- Java installed (JDK 17)
- Docker Desktop installed and running
- Maven Wrapper (mvnw) available in the project
- A valid pom.xml file

3. Building the JAR File
The Spring Boot application is first converted into an executable JAR file using Maven.

Command: ./mvnw clean package

Explanation: clean: Removes previous build artifacts, package: Compiles code and generates the JAR file

Output: The generated JAR file is located inside the target/ directory:
target/resume-0.0.1-SNAPSHOT.jar

Verification: The JAR file is tested locally before Dockerization:
java -jar target/resume-0.0.1-SNAPSHOT.jar

4. Dockerfile Creation
A Dockerfile is created to define how the backend application will run inside a container.

Dockerfile:
```
# Stage 1: Build the JAR
FROM maven:3.9.9-eclipse-temurin-17 AS build
WORKDIR /app
COPY . .
RUN mvn clean package -DskipTests

# Stage 2: Run the app
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY --from=build /app/target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

5. Dockerfile Explanation
#Stage 1
FROM maven:3.9.9-eclipse-temurin-17 AS build
Uses Apache Maven with Eclipse Temurin (Java 17)
Names this stage build.
WORKDIR /app
Sets /app as working directory inside container.
COPY . .
Copies entire Spring Boot project (source code, pom.xml, configs).
RUN mvn clean package -DskipTestsclean → removes old builds
package → compiles code and creates .jar
-DskipTests → skips test execution (faster build)
Output generated in: /app/target/<your-app-name>.jar

#Stage 2
FROM eclipse-temurin:17-jdk

Uses lightweight Eclipse Temurin (Java 17 runtime)
👉 No Maven here → smaller production image
WORKDIR /app
Sets working directory for application.
COPY --from=build /app/target/*.jar app.jar
Copies generated .jar from Stage 1
Renames it to app.jar
EXPOSE 8080
Exposes port 8080 (default for Spring Boot)
ENTRYPOINT ["java", "-jar", "app.jar"]
Starts the application using: java -jar app.jar

Stage 1 (Maven)Build .jar file
Stage 2 (Java Runtime)Run the application

6. Building the Docker Image
The Docker image is created using the Dockerfile.

Command: docker build -t resume-backend .

Explanation:
docker build: Builds the image
-t resume-backend: Tags the image with a name
. : Uses current directory as build context

7. Running the Docker Container
The Docker image is executed as a container.
Command: docker run -p 8080:8080 resume-backend

Explanation:
docker run: Starts a container
-p 8080:8080: Maps container port to host port
resume-backend: Image name

8. Application Access

Once the container is running, the backend application can be accessed at:
spring.data.mongodb.uri=mongodb://mongo:27017/biz

9. Key Benefits of Dockerization

- Ensures consistent environment across systems
- Eliminates dependency issues
- Simplifies deployment process
- Enables scalability and portability

10. Conclusion

The backend application is successfully containerized using Docker. The application can now run consistently across different environments using a single Docker command without requiring manual setup

📄 FRONTEND DOCKERIZATION DOCUMENTATION

1. Overview
The frontend of the Resume Project is built using Angular. The production build generates static files, which are served using Nginx inside a Docker container.

2. Prerequisites

- Node.js installed
- Angular project setup
- Docker installed and running

3. Building the Angular Application
Command:
npm install
npm run build

👉 Output is generated inside: dist/resume-frontend/browser/

4. Dockerfile Creation

```
# Stage 1: Build Angular
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build

# Stage 2: Serve with nginx
FROM nginx:alpine
# Copy built files from previous stage
COPY --from=build /app/dist/resume-frontend/browser /usr/share/nginx/html
# Fix Angular routing
COPY nginx.conf /etc/nginx/conf.d/default.conf
```

```
# Optional: rename index file if it exists
RUN if [ -f /usr/share/nginx/html/index.csr.html ]; then \
   mv /usr/share/nginx/html/index.csr.html /usr/share/nginx/html/index.html; \
   fi
EXPOSE 80
```

5. Dockerfile Explanation

#Stage 1

FROM node:18 AS build

Uses Node.js image to build the Angular app and names this stage build.

WORKDIR /app

Sets /app as working directory inside container.

COPY package.json ./*

Copies dependency files (package.json, package-lock.json) for installation.

RUN npm install

Installs all required Node modules.

COPY . .

Copies full Angular source code into container.

RUN npm run build

Builds Angular project into optimized production files inside:
dist/resume-frontend/browser

#Stage 2

FROM nginx:alpine

Uses lightweight Nginx image to serve static files.

COPY --from=build ...

Copies built Angular files from Stage 1 into Nginx web root: /usr/share/nginx/html

COPY nginx.conf ...

Adds custom Nginx configuration to handle Angular routing properly.

RUN (index rename logic)

Renames index.csr.html → index.html if present.

EXPOSE 80

Exposes HTTP port for accessing the application.

Stage 1 (Node.js) → Build Angular app
Stage 2 (Nginx) → Serve built static files

6. Building Docker Image

```
docker build -t resume-frontend .
```

7. Running Container

```
docker run -p 4200:80 resume-frontend
```

8. Application Access

http://localhost:4200

9. Key Benefits

- No Node.js needed in production
- Lightweight and fast
- Optimized static delivery

10. Conclusion

The Angular frontend is containerized and served efficiently using Nginx, ensuring high performance and portability.

📄 DATABASE (MONGODB) DOCUMENTATION

1. Overview
MongoDB is used as the database for storing resume data. It is run as a Docker container for easy setup and isolation.

2. Running MongoDB Container
docker run -d -p 27017:27017 --name mongo mongo

3. Explanation
-d → run in background
-p 27017:27017 → port mapping
mongo → official image

4. Access
Host: localhost
Port: 27017
Database: biz

5. Docker Networking (IMPORTANT)
Inside Docker:
Backend connects using → mongo:27017

Outside Docker:
Local app connects using → localhost:27017

6. Benefits
- No manual Mongo installation
- Easy setup and teardown
- Isolated environment

7. Conclusion

MongoDB runs as a containerized service and integrates seamlessly with backend using Docker networking.

DOCKER COMPOSE DOCUMENTATION (MOST IMPORTANT)

1. Overview
Docker Compose is used to manage multi-container applications including:

1. Frontend
2. Backend
3. MongoDB

2. docker-compose.yml

```
version: '3.8'

services:

mongo:
image: mongo
container_name: mongo
ports:
- "27017:27017"
volumes:
- mongo-data:/data/db

backend:
build: ./backend/resume
container_name: backend
ports:
- "8080:8080"
depends_on:
- mongo
environment:
- SPRING_PROFILES_ACTIVE=docker
volumes:
- ~/workspace/docs/resume:/app/uploads
```

```
frontend:
build: ./frontend/resume-frontend
container_name: frontend
ports:
- "4200:80"
depends_on:
- backend

volumes:
mongo-data:
```

3. Running Entire Application

```
docker-compose up --build
```

4. Stopping Application

```
docker-compose down
```

5. Advantages

- Single command startup
- Dependency management
- Clean environment setup

6. Architecture Flow

Browser → Frontend (Nginx) → Backend (Spring Boot) → MongoDB

7. Conclusion

Docker Compose simplifies deployment by orchestrating all services together, ensuring seamless communication and easy management.

Docker Desktop → Docker Hub → Deployment (Not till now)

1. Introduction

This document explains the complete workflow of:

- Building Docker images locally using Docker Desktop
- Pushing images to Docker Hub
- Pulling and running images on any system
- Running the application using Docker Compose

2. Core Concepts

Docker

Docker is a containerization platform that allows applications to run in isolated environments called containers.

🔹 **Docker Image**

- A **read-only template**
- Contains application code + dependencies

👉 Example:

- Backend image (Spring Boot)
- Frontend image (Angular + Nginx)

🔹 **Docker Container**

- Running instance of an image
- Executes the application

🔹 **Docker Hub**

- Cloud-based repository for storing Docker images
- Similar to GitHub, but for images

3. Local Setup (Docker Desktop)

🔹 **Build Images**

```
docker build -t resume-project-backend
docker build -t resume-project-frontend
```

🔹 **Verify Images**

```
docker images
```

🔹 **Run Containers**

```
docker run -d -p 8080:8080 resume-project-backend
docker run -d -p 4200:80 resume-project-frontend
```

4. Push Images to Docker Hub

🔹 **Login**

```
docker login
```

🔹 **Tag Images**

```
docker tag resume-project-backend thakurhritik7/resume-backend:v1
docker tag resume-project-frontend thakurhritik7/resume-frontend:v1
```

🔹 **Push Images**

```
docker push thakurhritik7/resume-backend:v1
docker push thakurhritik7/resume-frontend:v1
```

5. Pull Images from Docker Hub

On any system:

```
docker pull thakurhritik7/resume-backend:v1
docker pull thakurhritik7/resume-frontend:v1
```

6. Deployment using Docker Compose

🔹 **docker-compose.yml**

```yaml
version: '3.8'

services:

 mongo:
   image: mongo:latest
   ports:
     - "27017:27017"

 backend:
   image: thakurhritik7/resume-backend:v1
   ports:
     - "8080:8080"
   depends_on:
     - mongo
   environment:
     - SPRING_DATA_MONGODB_URI=${MONGO_URI}

 frontend:
   image: thakurhritik7/resume-frontend:v1
   ports:
     - "4200:80"
   depends_on:
     - backend
```

🔹 **.env**

MONGO_URI=mongodb://mongo:27017/resume_db

7. Run Application

docker-compose up -d

8. Access Application

- Frontend → http://localhost:4200
- Backend → http://localhost:8080

9. Stop Application

docker-compose down

10. Clean Run (Testing Cloud Pull)

Remove everything:

```
docker-compose down
docker rm -f $(docker ps -aq)
docker rmi -f $(docker images -q)
```

Run again:

docker-compose up -d

👉 Images will be automatically pulled from Docker Hub

11. Important Learnings

🔹 **Image Pull Logic**

If image exists locally → use it
If not → pull from Docker Hub

🔹 **Container Communication**

Containers communicate using **service names**, not localhost:

- `backend → mongo`
- `frontend → backend`

🔹 **Versioning**

Use tags: v1, v2, v3

Avoid: latest ❌

12. Common Mistakes

- Using `localhost` inside containers
- Not tagging images properly
- Forgetting `docker login`
- Deleting volumes (data loss)
- Using only `latest` tag

13. Outcome

After completing this setup:

- The application becomes **portable**

- Runs on any system with Docker
- No manual setup required
- One-command deployment

14. Future Enhancements

- Deploy on VPS (Hostinger / AWS)
- Add domain and HTTPS

15. Conclusion

This workflow demonstrates a complete DevOps pipeline:

Local Development → Docker Images → Docker Hub → Deployment via Compose

Final Line

👉 *"I containerized my full-stack application, pushed versioned images to Docker Hub, and created a Docker Compose-based deployment system that allows anyone to run the application with a single command."*